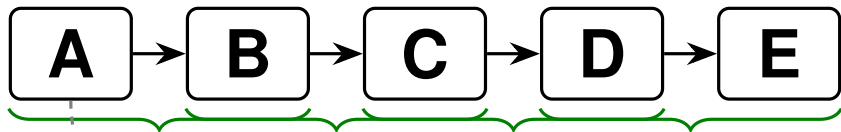


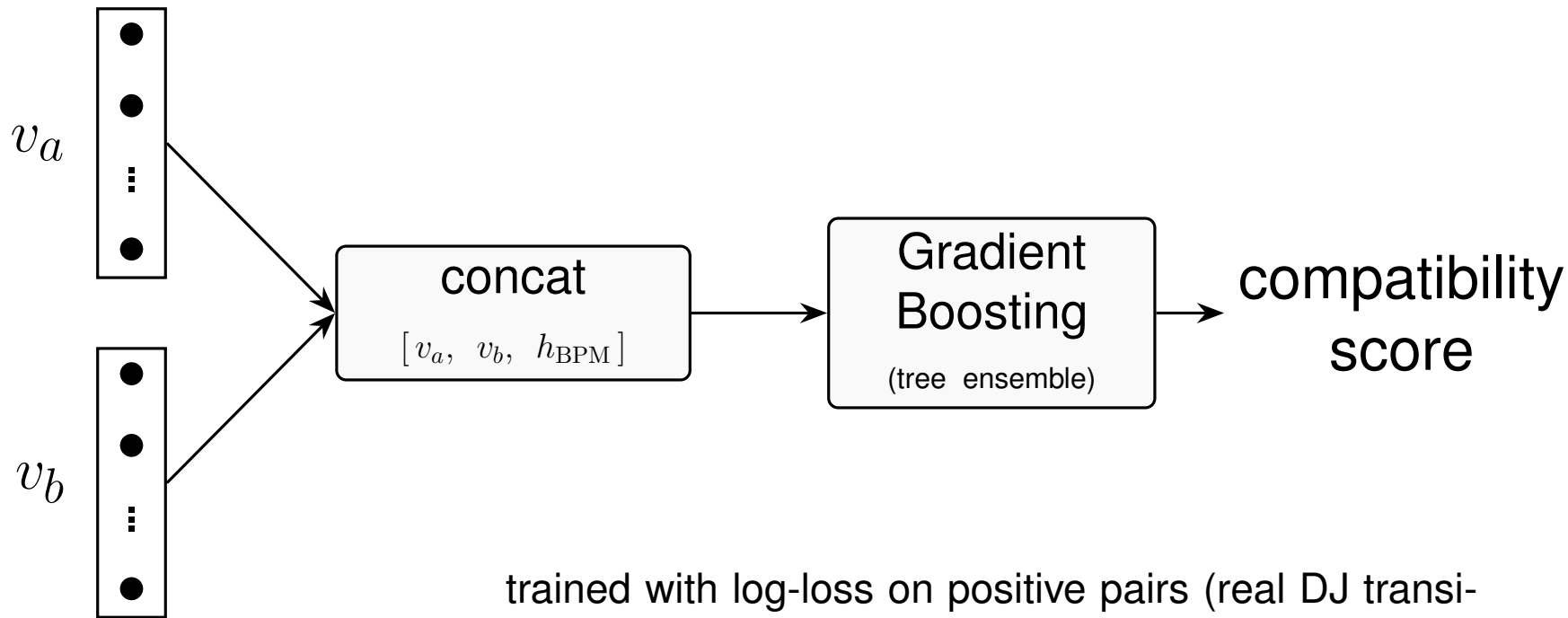
Mix #0421



positive pairs



negative pair
(Z sampled randomly from a different mix)



trained with log-loss on positive pairs (real DJ transitions) and negative pairs (random non-adjacent tracks)

80/20 split by mix to prevent leakage

$$\text{HBPM}(T_A, T_B) = \min(|T_A - T_B|, |T_A - 2T_B|, |T_A - T_B/2|)$$

Three rhythmic relationships

	Match	Double-time	Half-time
T_A (BPM)	128	140	90
T_B (BPM)	128	70	180
Plain $ T_A - T_B $	0	70	90
Harmonic HBPM	0	0	0

DJs mix at matched tempo, double-time, or half-time. Harmonic BPM takes the minimum distance across all three relationships, so a 90 BPM hip-hop track and a 180 BPM drum-and-bass track are scored as tempo-compatible.



raw audio



librosa
feature extraction
→ wide feature vector



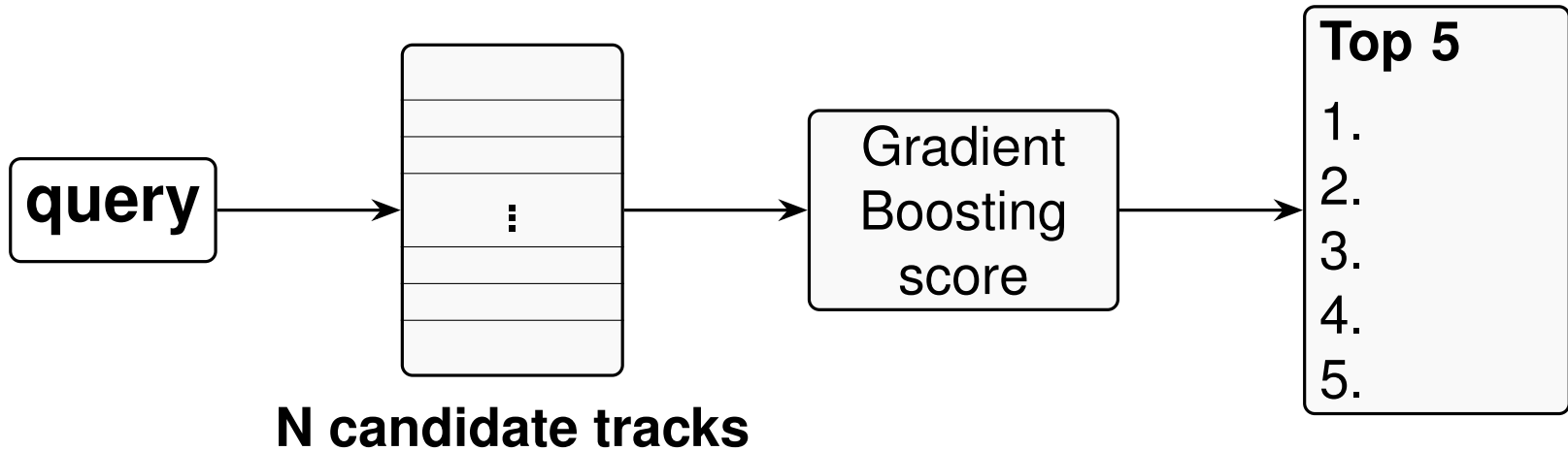
append
harmonic BPM
engineered feature



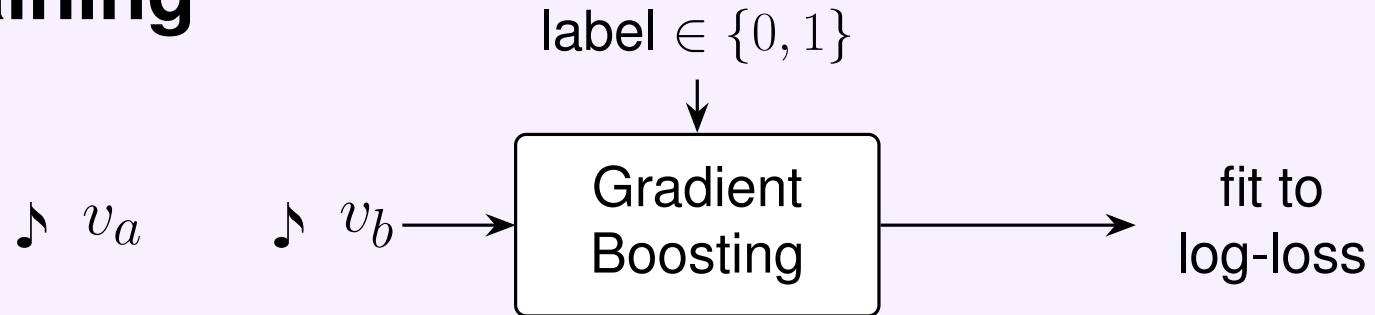
Gradient
Boosting
trained on
+ and - pairs



inference:
score query vs.
all candidates,
return top 5

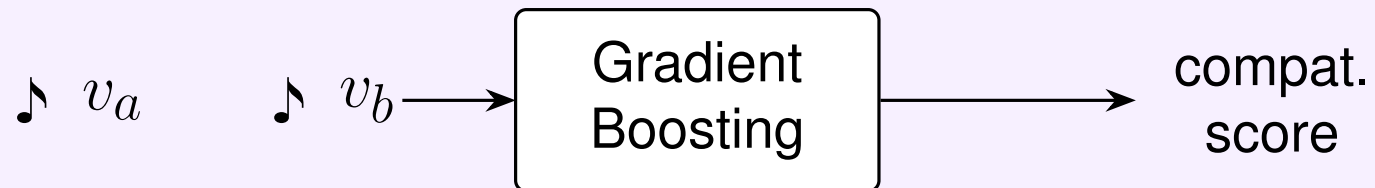


Training



5,940 pair examples from 80% of mixes
positives = real DJ transitions | negatives = random non-adjacent tracks

Testing



1,502 held-out pairs from the 20% of mixes never seen during training
pair classification: **77.0%** accuracy | retrieval: **Hit@5 = 1.47%**

(747 queries scored against all 3,827 candidate tracks; MRR = 0.014)